Advanced Defence Q&A

LoRA-Based Resource-Efficient LLM Adaptation for RTL Hardware Trojan Detection

Rule for hard questions: Never become defensive. State the evidence, acknowledge the limitation, and explain the scientifically correct next experiment.

1. Dataset Leakage and Experimental Validity

Q: You call the evaluation leakage-safe. How can you claim that when synthetic generation may leave recognizable patterns?
Defence answer: The precise claim is that we removed *explicit label leakage*: temporary `HT_START`, `HT_END`, Trojan-specific comments, and direct annotations were removed from final prompts. However, synthetic insertion can still create structural artifacts. Therefore, the results demonstrate label-sanitized performance in a controlled setting, not complete immunity to all dataset leakage.

Q: Could the model simply detect your Trojan template instead of detecting a general Hardware Trojan?
Defence answer: That is a valid threat to external validity. The experiment proves that the model can detect controlled Trojan-inserted RTL patterns after explicit labels are removed. It does not prove general detection of every real attacker-designed Trojan. Template-separated and external-benchmark evaluation are necessary next steps.

Q: Why did you not enforce template-level separation between training and test data?
Defence answer: The current study used a randomized stratified split to preserve class balance and provide a controlled first evaluation. The paper explicitly identifies the absence of template-level separation as a limitation. A stronger future protocol would group related RTL designs and insertion templates into separate partitions.

Q: If similar RTL modules exist in training and testing, is your reported accuracy inflated?
Defence answer: It may be optimistic because structural similarity across splits cannot be ruled out. That is why I present 98.89% as controlled randomized-test performance rather than a universal generalization result.

Q: Why should we trust a 90-sample test set?
Defence answer: The test set gives a clear controlled result: 89 correct predictions out of 90. However, 90 samples is still modest for broad statistical claims. Larger held-out benchmarks, confidence intervals, repeated random seeds, and grouped cross-validation would strengthen confidence.

Q: Why did you not report confidence intervals?
Defence answer: The paper reports point estimates. Confidence intervals would be valuable because they show uncertainty caused by finite test-set size. This should be included in a more extensive follow-up study.

Q: Did you check for duplicate RTL files across data splits?
Defence answer: The paper does not report a duplicate-detection or near-duplicate analysis, so I would not claim that it was performed. In future work, I would use code similarity analysis and design-family grouping before splitting.

Q: Can preprocessing itself create a new leakage signal?
Defence answer: Yes. Any deterministic preprocessing pipeline can introduce regularities. The role of normalization and obfuscation is to reduce superficial shortcuts, but only robust ablations and external testing can show whether those shortcuts have truly been reduced.

2. Model Choice and LoRA

Q: Why use an LLM for this task instead of a graph neural network or a traditional classifier?
Defence answer: RTL contains code-like sequential structure, dependencies, conditions, and assignments that code-capable transformers can model from serialized text. Graph neural networks are also a strong alternative because circuits naturally form graphs. A fair comparison would require the same dataset and evaluation protocol, and this is a valuable future baseline.

Q: Why Qwen2.5-1.5B-Instruct specifically?
Defence answer: It offers a reasonable balance of language-model capacity and practical resource use. The study targets low-resource academic experimentation, so a 1.5-billion-parameter instruction model is more feasible than much larger models.

Q: Why did you not use a hardware-specialized pretrained model?
Defence answer: The selected model provides a practical instruction-tuned base for the controlled experiment. A hardware-specialized model could be an important comparison in future work, particularly to test whether domain-specific pretraining improves generalization.

Q: What is the mathematical intuition behind LoRA?
Defence answer: Instead of learning a full weight update ΔW, LoRA approximates it using low-rank matrices:

$$\Delta W = BA,$$

where A and B have much smaller rank than the original weight matrix. The adapted weight becomes $W' = W + BA$, while the pretrained W remains frozen.

Q: Why does LoRA reduce memory use?
Defence answer: Full fine-tuning stores gradients and optimizer states for nearly all model parameters. LoRA only optimizes small adapter matrices, so the training-state memory requirement is much lower.

Q: Does LoRA guarantee the same performance as full fine-tuning?
Defence answer: No. LoRA is a trade-off between adaptation capacity and resource efficiency. It can perform very well, but full fine-tuning may outperform it in some tasks or datasets at a substantially higher computational cost.

Q: Why did you choose LoRA rank 8?
Defence answer: Rank 8 was the reported configuration for this low-resource prototype. It gives a compact adapter capacity. A complete study should include rank ablations, for example ranks 4, 8, 16, and 32, to quantify the accuracy-resource trade-off.

Q: Which exact layers were targeted by LoRA adapters?
Defence answer: The accepted paper reports LoRA rank, the PEFT strategy, and trainable parameter count, but it does not document the target transformer modules in detail. I would not claim a specific target-module configuration without implementation evidence.

Q: Why use FP16 rather than INT8 or 4-bit quantization?
Defence answer: The paper notes exploratory QLoRA-style low-bit adaptation, but the final reported workflow used FP16 LoRA adaptation for runtime stability and reproducibility. Low-bit quantization can reduce memory further, but it can also introduce implementation and stability considerations.

Q: Why train for only three epochs?
Defence answer: Three epochs were part of the reported resource-constrained configuration. More epochs could improve task fit or could overfit a small synthetic dataset. A validation-based learning-curve study would provide a stronger justification.

3. Detection Metrics and Localization

Q: Your accuracy is high. Could it be misleading?
Defence answer: Yes. Accuracy can be misleading in an imbalanced dataset, although our test set is balanced with 45 clean and 45 Trojan samples. That is why we also report precision, recall, F1-score, the confusion matrix, TLC, and PLC.

Q: Why is precision 100% while recall is lower?
Defence answer: The model produced no false-positive Trojan predictions, so Trojan-class precision is 100%. It missed one actual Trojan sample, so recall is $44/45 = 97.78\%$.

Q: Which is more dangerous: a false positive or a false negative?
Defence answer: A false negative can allow a Trojan to remain undetected, which is serious in a security context. A false positive increases verification cost and may waste engineer time. The preferred trade-off depends on the security-critical application.

Q: Why did you not optimize for recall if missed Trojans are dangerous?
Defence answer: The reported results are from the selected classification configuration rather than an explicitly recall-optimized threshold study. A deployment-oriented system should assess threshold tuning, cost-sensitive learning, and the acceptable false-negative rate for the target threat model.

Q: What exactly does TLC measure?
Defence answer: TLC measures trigger-node coverage:

$$\mathrm{TLC} = \frac{|T_p \cap T_g|}{|T_g|},$$

where T_p is the predicted trigger-node set and T_g is the ground-truth trigger-node set. It measures how much of the actual trigger region is recovered.

Q: What exactly does PLC measure?
Defence answer: PLC measures payload-node coverage:

$$\mathrm{PLC} = \frac{|P_p \cap P_g|}{|P_g|},$$

where P_p is the predicted payload-node set and P_g is the ground-truth payload-node set.

Q: Can TLC and PLC be high if the model predicts many irrelevant nodes?
Defence answer: Yes. TLC and PLC are coverage-oriented metrics. They reward recovering ground-truth nodes but do not strongly penalize extra predicted nodes. Future work should add localization precision, region-level F1-score, exact-match rate, and intersection-over-union.

Q: Why is localization ground truth reliable in this study?
Defence answer: Ground truth is reliable with respect to the controlled insertion procedure because the system records the inserted trigger and payload regions as metadata. This reliability applies to the synthetic experiment, not necessarily to naturally occurring or unknown real-world Trojans.

Q: Does high localization coverage prove that the model understands circuit semantics?
Defence answer: No. High coverage demonstrates agreement with recorded synthetic ground truth in the evaluation setup. Semantic understanding requires stronger evidence, such as testing on novel design families, novel Trojan types, and independently labeled real-world benchmarks.

Q: How does the model output a localization prediction?
Defence answer: The paper evaluates predicted suspicious RTL regions against recorded trigger and payload metadata. The paper does not provide a complete formal specification of the output parser or region-decoding mechanism, so I would not claim details beyond the reported methodology.

4. Threat Model and Security Questions

Q: What is your exact threat model?
Defence answer: The work assumes controlled RTL-level insertion of trigger and payload logic into clean Verilog modules. It evaluates whether an adapted model can detect and localize these modifications after explicit annotations are removed.

Q: Can your method detect a Trojan inserted by a sophisticated adversary who knows your training method?
Defence answer: Not yet established. An adaptive adversary could use structurally different logic, rare coding styles, delayed activation, or adversarial transformations. Robustness against adaptive attackers requires dedicated adversarial evaluation.

Q: Can an attacker evade the model by renaming signals or reformatting code?
Defence answer: Renaming and formatting transformations are relevant concerns. The preprocessing pipeline attempts to reduce dependence on superficial textual artifacts, but systematic robustness testing against semantics-preserving transformations is needed.

Q: Can this method detect dormant Trojans that never activate in simulation?
Defence answer: Potentially, RTL analysis can inspect suspicious structural patterns without requiring runtime activation. However, detection performance for arbitrary dormant Trojans is not established by this controlled dataset alone.

Q: Can this approach detect analog Trojans, doping Trojans, or layout-level attacks?
Defence answer: No. The current work is limited to RTL text representations. Analog, doping, gate-level, and physical-layout Trojans require different data, representations, and security-analysis techniques.

Q: Can your system formally prove that a chip is Trojan-free?
Defence answer: No. This is a learned detection and localization aid, not a formal proof system. Formal verification, equivalence checking, security review, and physical testing remain necessary for high-assurance hardware security.

Q: Could the LLM hallucinate a Trojan location?
Defence answer: Yes. LLM-generated predictions can contain incorrect or overconfident outputs. Therefore, the system should prioritize regions for expert review or downstream formal checks rather than make an autonomous final security decision.

5. Comparison and Reproducibility

Q: Why did you not directly compare with PEARL?
Defence answer: PEARL uses a different model, adaptation strategy, hardware environment, dataset, and evaluation setting. Comparing final accuracy values without matching those conditions would be scientifically weak. We use PEARL as contextual prior work, not as a directly matched benchmark.

Q: Does your result outperform PEARL?
Defence answer: I do not claim that. Our work emphasizes low-resource LoRA-based experimentation, whereas PEARL has a different research focus and experimental setup.

Q: Why did you not compare against a zero-shot LLM baseline?
Defence answer: That would be a useful ablation. The accepted paper focuses on the controlled LoRA-based framework and does not report a zero-shot baseline. A stronger future study should compare zero-shot, few-shot, LoRA, full fine-tuning, and classical hardware-security baselines.

Q: Why did you not compare against traditional ML or GNN models?
Defence answer: A matched comparison would make the study stronger. It requires identical data splits, preprocessing, and evaluation metrics to be meaningful. This is a clear future-work direction.

Q: Can another researcher reproduce your six-minute runtime?

Defence answer: They may obtain a similar order of runtime under comparable conditions, but exact runtime can vary due to GPU allocation, Colab environment, software versions, initialization, and data loading. The reported value is approximate, not a guaranteed constant.

Q: Is Google Colab suitable for reproducible research?
Defence answer: It is useful for accessible experimentation, but reproducibility should also document package versions, random seeds, data-processing code, model configuration, and exact training commands. Cloud hardware availability can vary.

Q: What would a complete reproducibility package contain?
Defence answer: It should contain the clean-data source references, controlled Trojan-generation code, preprocessing and sanitization code, split files, random seeds, LoRA configuration, training scripts, inference and localization parser, metric code, and environment specifications.

6. Strong Closing Answers

Q: What is the one sentence contribution of your paper?
Defence answer: We present a controlled, resource-efficient RTL Trojan detection and localization workflow using LoRA adaptation of Qwen2.5-1.5B-Instruct, evaluated on a balanced synthetic RTL dataset with explicit limitations stated.

Q: What is the one sentence limitation of your paper?
Defence answer: The strong results come from a controlled synthetic RTL setting with a randomized split, so they do not yet prove generalization to structurally novel, real-world, gate-level, or physical-design Trojans.

Q: What is the most important future experiment?
Defence answer: A template-separated evaluation on larger and heterogeneous real-world RTL Trojan benchmarks, with matched classical, graph-based, zero-shot, and PEFT baselines.

Q: Why is this work still useful despite its limitations?
Defence answer: It establishes a reproducible starting point for studying LLM-based RTL security analysis under limited hardware resources. It also exposes the exact validation gaps that future work must address before deployment claims are justified.